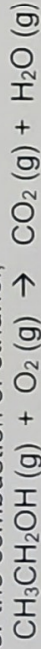


3. Calculate the enthalpy change for the combustion of ethanol,

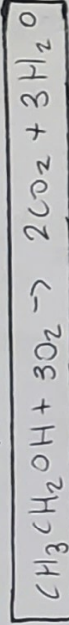


given the following bond energies: (4 MARKS)

NOTE: THE EQUATION IS NOT BALANCED!

GRASP not required.

To balance

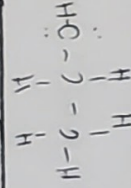


(6 x 10)

x³

(H²O)

$\Delta H = \text{Bonds Broken} - \text{Bonds Formed}$



C-H	5 (414)
C-C	1 (347)
C-O	1 (360)
O-H	1 (464)
O=O	3 (498)

C=O	2 (799) x 2
O-H	6 (464)

$$\Delta H = 4735 \text{ kJ/mol} - 5980 \text{ kJ/mol} = -1245 \text{ kJ/mol}$$

COMMUNICATION (10 marks total)

e. Change in H⁺ is

$$\Delta H = -1245 \text{ kJ/mol}$$

3.5

1. For a molecule of C₂H₄, complete the following: (4 marks total)

(a) Draw the Lewis structure. (1 mark)

4

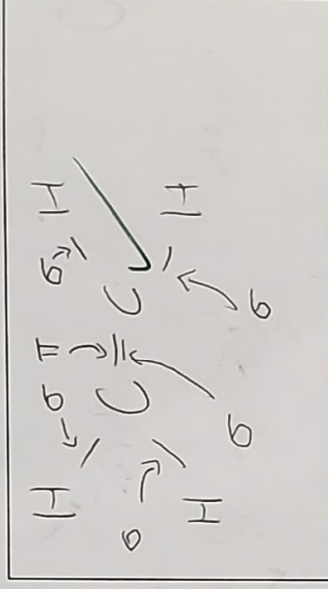
(b) What is the hybridization of the central carbon atom?

sp² (1 mark)

AX³

(c) Predict the VSEPR shape. Trigonal planar (1 mark)

(d) Identify any sigma or pi bonds by labelling them on your Lewis structure that you drew in part (a). (1 mark)



2. Compare TeCl₄ and CCl₄ on the various aspects of molecular shape and related properties given and choose the letter below that matches the property given. Letters can be used twice. (3 marks - ½ mark each)

(a) if the answer is only TeCl₄

(b) If the answer is only CCl₄

(c) If the answer is both TeCl₄ and CCl₄

(d) If the answer is neither TeCl₄ nor CCl₄

a ✓ i. Polar molecule

b ✓ ii. Symmetrical shape

a ✓ iii. Contains lone pairs

b ✓ iv. Has a larger bond angle

b ✓ v. Would not dissolve in water

d ✓ vi. The presence of double or triple bonds

Like dissolves like

3. What is wrong with each of the following notations? Briefly explain the error. (2 marks)

a) (3, 3, 0, -1/2)

For $n=3$, the ℓ block

has no orbitals ✓

$l \neq 3$, only can equal

0, 1, or 2 when $n=3$

b) (4, 2, 0, 0)

The spin m_s cannot have a value of 0, only $-\frac{1}{2}$ and $+\frac{1}{2}$

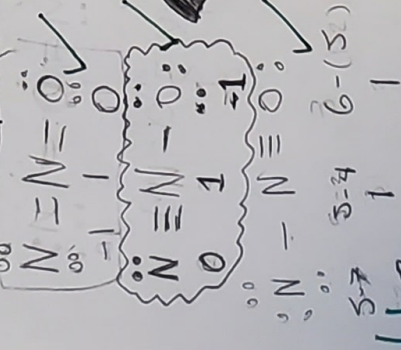
$$m_s = \pm \frac{1}{2}$$

4. According to hybrid orbital theory, why can PCl_5 exist but NCl_5 cannot? (1 mark)

The reason PCl_5 exists is because P is able to exceed the octet rule while N is not able to. Cl_3PF_2 are all bonded to the octet rule, so NCl_5 is invalid. (Using hybrid orbital theory we see sp^3d is the required hybridization, since N has to fill 3s before the d orbital

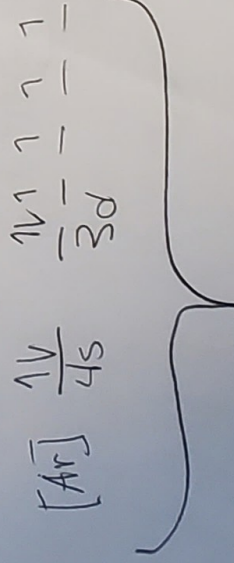
it cannot hybridize using d orbitals, so the first 2 rows cannot exceed the octet rule and that includes N.

APPLICATION (10 marks total) Draw three reasonable Lewis structures for N_2O (hint: N is the central atom). Use formal charge to indicate the most probable structure and circle it. (4 marks)



Most probable ✓ The most probable structure is $\text{N} \equiv \text{N} - \text{O}$ because -1 is on the most electronegative atom and all formal charges are as close to 0 as they can.

Some transition metals such as iron (2^+ and 3^+) have two possible charges; use a partial energy level diagram (4s and 3d orbitals only) to explain how this is possible. (2 marks)



ground state of $[\text{Fe}]$

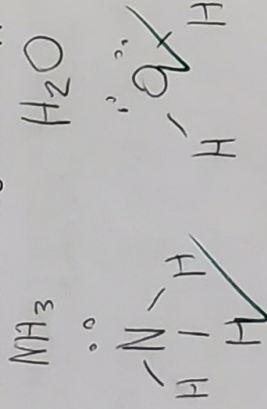
In Iron^{2+} the 4s shell is empty so it is a possible state since those 7 electrons are for the nucleus.

$$\text{Fe}^{2+} [\text{Ar}] \quad \begin{array}{c} 11 \\ 4s \end{array} \quad \begin{array}{c} 11 \\ 3d \end{array}$$

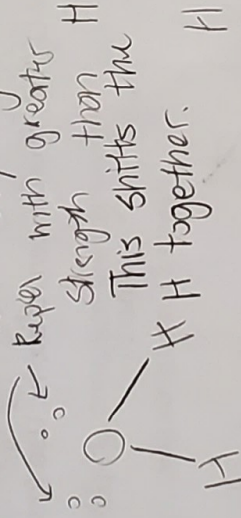
In Iron^{3+} it is possible since the last electron in 3d is loosely held as the 3d ionization energy is low, allowing it to lose 3 electrons.

$$\text{Fe}^{3+} [\text{Ar}] \quad \begin{array}{c} 11 \\ 4s \end{array} \quad \begin{array}{c} 11 \\ 3d \end{array}$$

3. Why are the bond angles in a molecule of NH_3 (107.3°) larger than the angles in a molecule of water (104.4°)? Draw Lewis diagrams to support your answer. (4 marks)



The reason NH_3 has larger bond angles even though they are both tetrahedral (4 groups on central atom), is because of the amount of lone pairs. H_2O has 2 lone pairs, since the lone pairs have a stronger repulsion force from each other compared with the Hydrogens bonded on the O.



When comparing to NH_3 there is only one lone pair on N, so the angle is less than 109.5° but

does not repel as much as two lone pairs since the negative charges repel (Coulomb's law) more if there are several.

Answer

1. For each of the following molecules, do the following: (12 marks)

- | Compound | Lewis Structure | VSER Shape | Polar/Non-Polar | Hybridization Central |
|------------------|-----------------|------------------------|--|-----------------------|
| SeO ₂ | | Trigonal planar
X | Polar ✓ | sp ² ✓ |
| SbH ₃ | | Tetrahedral
X | Polar ✓
(be asymmetric + lone pair) | sp ³ ✓ |
| PF ₅ | | Trigonal Bipyramidal ✓ | Non-polar ✓ | sp ³ d ✓ |

- 3
-2
- $[Kr] \frac{1\cancel{v}}{5s^2} \frac{1\cancel{v}1\cancel{v}1\cancel{v}1\cancel{v}}{4d^{10}} \frac{1\cancel{v}1\cancel{v}}{5p^4}$ ground state
- $[Kr] \frac{1\cancel{v}}{5s^2} \frac{1\cancel{v}1\cancel{v}1\cancel{v}1\cancel{v}}{4d^{10}} \frac{1\cancel{v}1\cancel{v}1\cancel{v}}{5p^4}$ Remove from last electrons.
- +6
- $[Kr] \frac{1\cancel{v}1\cancel{v}1\cancel{v}1\cancel{v}}{4d^{10}}$

- 51